Security and privacy protection:

All students personal data including lifestyle preferences, medical/disability information, nationality, and results that are processed by the allocation engine shall be encrypted both in transit (TLS 1.3 or higher) and at rest (AES-256).

Access to the automated allocation function and its results shall be restricted to authorised Accommodation Office staff only, enforced though role-based access control(RBAC) integrated with the university single sign-on (SSO). when it comes to lifestyle and compatibility data by the AI engine shall be anonymised whenever possible before processing. Full identifiable data should only be accessible on a strict need to know basis. Also the system should maintain a complete, immutable audit log of every allocation run, including who initiated it, parameters used, and final assignments. Logs shall be retained for a minimum of 7 years in compliance with UK data protection and university retention policies.

The allocation engine shall comply with the uk GDPR and the Data protection Act 2018. Students shall be able to request explanation of allocation decisions that affect them. Any integration with external service providers shall ensure that personal data is not transferred outside the UK without appropriate safeguards.

Performance:

A full automated allocation run for up to 5000 approved applicants and 6000 rooms shall complete within 10 minutes under normal load. During peak periods for example if it were results day, the system shall support at least 50 concurrent Accommodation Office staff users viewing or triggering allocation processes without degradation of response time below 3 seconds for standard queries. Individual room allocation queries or manual override checks shall return results in around a second.

Reliability:
The automated allocation service shall have a minimum availability of 99.5%
During the core allocation periods(August-October) furthermore the system shall guarantee that an allocation run either completes successfully with a consistent set of assignments or fails cleanly without partial or corrupted allocations.

In the event of a failure, for example if there was to be an AI engine timeout, or if the database was to be unavailable the system will automatically roll back any partial changes and notify the staff member with a clear error message within 30 seconds.

The allocation results shall be reproducible: running the engine with identical input data should give out the same output.

The Mean Time To Recovery (MTTR) for the allocation microservice shall not exceed 15 minutes.

Scalability:

The allocation engine shall scale to support at least 3x increase in applicants and rooms without requiring redesign and also the original design will also support multi-tenant operation so that allocation workloads from different university tenants do not interfere with each other. The service shall automatically scale compute resources up and down based on demand to handle sudden spikes such as results day while remaining cost-efficient during quiet periods.